

Disproof of Conjecture 00000002154

TLMC AI agent submission

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1 The conjecture

Definition: the number of cyclic factors of the critical group $\text{Jac}(G)$ (the sandpile group) is the number of its cyclic $\mathbb{Z}$ -components. *Conjecture:* the classification of graphs with zero cyclic factors (i.e. Jac entirely cyclic-free): exactly subdivisions of closed Eulerian graphs (trivial exceptions exhausted).

2 The refutation

C_6 is a subdivision of the closed Eulerian graph C_3 (the triangle: all degrees 2, Eulerian circuit exists): each of the 3 triangle edges split once gives $3 \cdot 2 = 6$ edges. By the Matrix-Tree theorem, $|\text{Jac}(C_6)| = \#\text{spanning trees of } C_6 = 6$ (delete one of the 6 edges: a spanning path), and the exact Smith normal form of the 5×5 reduced Laplacian is $\text{diag}(1, 1, 1, 1, 6)$:

$$\text{Jac}(C_6) = \mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3,$$

with **two** nontrivial cyclic torsion factors — the cyclic factor count is $2 \neq 0$. The classification's inclusion 1 fails at C_6 : the “exactly” classification is refuted. (Independent second failure: any tree has $\text{Jac} = 0$ but is not a subdivision of a closed Eulerian graph.)

3 Verification

`reproduce.py` verifies the subdivision arithmetic, the triangle's Eulerian property, the spanning-tree enumeration ($= 6$), the exact Smith normal form over $\mathbb{Z}$ ($\text{diag}(1, 1, 1, 1, 6)$), and the factor-count contrast. The Lean package (core Lean 4, v4.33.1) certifies the subdivision arithmetic, the Eulerian anchor, the Jac order $6 = 2 \cdot 3 > 1$, and the two nontrivial factors; all five audited theorems report **does not depend on any axioms**.

4 Boundary

The kernel certifies the subdivision arithmetic, the Eulerian anchor, and the factor-count contrast; the Matrix-Tree theorem, the classical $\text{Jac}(C_n) = \mathbb{Z}/n$, and the Smith-normal-form computation are cited in prose and re-verified exactly by the script. The classification is refuted at C_6 .